

$$1 - \operatorname{sech}^2 t = \tanh^2 t. \quad [3]$$

[illegible]
$$x = \frac{1}{2} \tanh^2 t + \ln \operatorname{sech} t, \quad y = 1 + \tanh^4 t, \quad \text{for } t > 0.$$

(b) Show that $\frac{dy}{dx} = -4\operatorname{sech}^2 t$. [5]

This image shows a full page of white paper with horizontal dashed lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

- (c) Find the coordinates of the point on C with $\frac{d^2y}{dx^2} = -\frac{9}{2}$, giving your answer in the form $(a + \ln b, c)$ where a, b and c are rational numbers. [6]

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.